

Hongbin Zhong

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🌐 rjzhb

Education

Georgia Institute of Technology
Ph.D. in Computer Science

Aug 2024–2028 (*expected*)

Northeastern University
B.S. in Computer Science

2020–2024

Work Experience

Microsoft Research

Redmond, WA | May 2025 – Aug 2025

LLM Agent Research Intern

Mentors: Adriana Szekeres, Suman Nath

- Proposed a **memory-augmented WebAgent** with an **updatable state-machine graph** of web states and UI/data verbs, enabling deterministic graph traversal instead of fragile step-by-step visual guessing.
- Designed a **program-based planner** that synthesizes a complete executable Python program in **one inference step** (avg **1 LLM call/task**), improving WebArena success to **90.6%** and reducing avg time to **62s**.
- Implemented **dynamic adaptation**: trigger vision-based recovery on runtime exceptions and write back fixes to the memory graph for continual robustness under evolving UIs.

4Paradigm (Top AI company in China, Series-D Unicorn)

(Remote) | Feb 2024 – Apr 2024

AI Engineer (Part-Time)

- Optimized the **LLM inference** for an AI assistant, redesigning **KV-cache eviction** to cut **end-to-end latency**.

4Paradigm (Top AI company in China, Series-D Unicorn)

Beijing | Jan 2023 – Jul 2023

AI System Research Intern

- Optimized **online ML feature store**, focusing on **low-latency** feature joins under **late / out-of-order events** (training-serving consistency).
- Designed **PECJ** (proactive error-compensation) to **bound feature staleness/incorrectness** while preserving latency; achieved up to **70% lower error at the same latency** and integrated into the **multi-threaded** streaming feature pipeline (github.com/4paradigm/OpenMLDB 🔗, ★ **1.7k stars**).

Meituan, Top 5 Chinese Internet Company Serving 600M+ Users

Beijing, China | Apr 2022 – Sep 2022

Software Engineer Intern

- Shipped foundational backend for the Meituan short-video product, supporting new content formats.
- Built Kafka/Hive reporting pipelines to feed **recommendation models** and periodic refresh logic for low-bandwidth users.

Project Experience

InfiniFlow (Vector Database)

🔗 ★ 4.2k stars

Remote | Mar 2024 – Apr 2024

Contributor, Storage & Indexing Optimization

- Decoupled GC watermark tracking from the heavy `txn_map` to a lightweight **Red-Black Tree** (`std::set`), optimizing critical-path lookup complexity from linear scanning to $O(1)$.
- Replaced fine-grained locking with a **batched object deletion** protocol, significantly reducing **mutex contention** on the Buffer Manager and amortizing file I/O syscall overheads.

Long-Context LLM Inference Acceleration

Atlanta, GA · | Dec 2025 – PRESENT

KMeans-based KV Compression for Fast Long-Context Decoding

- Implemented a **KMeans-based KV compression/indexing** layer for long-context decoding, using **centroid summaries** to reduce KV reads/attention compute while maintaining output quality.
- Added two practical decoding tricks: **dynamic top-k** (adaptive retrieval budget per head/task instead of fixed k) and a lightweight **window cache** (always keep a small prefix/suffix window and prioritize it during retrieval); optimized the decode path with custom **CUDA** kernels and online updates, achieving **6×** attention speedups with **single-digit** clustering overhead.

Quantized LLM Inference Kernel Optimization

Atlanta, GA · | Feb 2025 – PRESENT

Fused INT4 Dequant + GEMM for Small-batch LLM Serving (Triton/CUDA)

- Built a **fused INT4 dequant + GEMM** kernel for small-batch LLM inference to reduce memory traffic and kernel launch overhead, targeting memory-bound decode shapes.
- Added **SplitK-style parallelism** to improve SM utilization under skinny GEMMs; on an **A40** GPU, **expected ~1.3–1.7×** speedup on representative decode workloads (shape-dependent), compared to a non-fused baseline. (*inspired by published work; engineering implementation + profiling/tuning are mine*)

Academic Papers

- [**AI Agent**] Beyond Screenshots: A Dynamic State-Machine Memory and Global Programmatic Planner for Web Agents
Hongbin Zhong* with Microsoft Research collaborators
Achieved about 90% accuracy on the WebArena benchmark, surpassing all existing SOTA methods; results to be released soon.
- [**Efficient RAG**] HoneyBee: Efficient Role-based Access Control for Vector Databases via Dynamic Partitioning
Hongbin Zhong, Matthew Lentz, Nina Narodytska, Adriana Szekeres, Kexin Rong
SIGMOD 2026
- [**Efficient AI**] StreamFlow: Theory, Algorithm, and Implementation for High-Efficiency Rectified Flow Generation
Sen Fang*, **Hongbin Zhong***, Yalin Feng, Dimitris N. Metaxas
submitted to ICML
- [**Efficient AI**] PECJ: Stream Window Join on Disorder Data Streams with Proactive Error Compensation
Xianzhi Zeng, Shuhao Zhang, Hongbin Zhong, Hao Zhang, Mian Lu, Zhao Zheng, Yuqiang Chen
SIGMOD 2024
- [**Data Analytics**] FaDE: More Than a Million What-ifs Per Second
Haneen Mohammed*, Alexander Yao*, Charlie Summers*, Hongbin Zhong, Gromit Yeuk-Yin Chan, Subrata Mitra, Lampros Flokas, Eugene Wu
VLDB 2025
- [**Diffusion model**] Stable Signer: Hierarchical Sign Language Generative Model
Sen Fang*, Yalin Feng*, Hongbin Zhong, Yanxin Zhang, Dimitris N. Metaxas
submitted to ACL
- [**Diffusion model**] Signdiff: Diffusion model for american sign language production
Sen Fang; Chunyu Sui; Yanghao Zhou; Xuedong Zhang; Hongbin Zhong; Yapeng Tian
submitted to CVPR

Technologies

Languages: C++, C, Python, Java, SQL, Rust

Proficiency: Efficient AI, CUDA Programming, AI compiler, AI Agent, RL, Storage